

# LAB 3

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Introduction to Image Generation

# HKUST API

| OpenAI Model name | Azure Deployment name | Model version    | API version |
|-------------------|-----------------------|------------------|-------------|
| dall-e-3          | dall-e-3              | 3.0              |             |
| gpt-35-turbo      | gpt-35-turbo          | 0125             |             |
| gpt-35-turbo-16k  | gpt-35-turbo-16k      | 0613             |             |
| gpt-4-turbo       | gpt-4-turbo           | turbo-2024-04-09 |             |
| gpt-4.1-mini      | gpt-4.1-mini          | 2025-04-14       |             |
| gpt-4o            | gpt-4o                | 2024-08-06       |             |
| gpt-4o-mini       | gpt-4o-mini           | 2024-07-18       |             |

|            |            |            |                    |
|------------|------------|------------|--------------------|
| gpt-5-mini | gpt-5-mini | 2025-08-07 |                    |
| gpt-5-nano | gpt-5-nano | 2025-08-07 |                    |
| o1-mini    | o1-mini    | 2024-09-21 | 2025-02-01-preview |
| o3-mini    | o3-mini    | 2025-01-31 | 2025-02-01-preview |
| o4-mini    | o4-mini    | 2025-04-16 | 2025-02-01-preview |

# HKUST API

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|-------------------|-----------------------|---------------|--------------------|
| gpt-5-mini        | gpt-5-mini            | 2025-08-07    |                    |
| gpt-5-nano        | gpt-5-nano            | 2025-08-07    |                    |
| o1-mini           | o1-mini               | 2024-09-21    | 2025-02-01-preview |
| o3-mini           | o3-mini               | 2025-01-31    | 2025-02-01-preview |
| o4-mini           | o4-mini               | 2025-04-16    | 2025-02-01-preview |

```
from openai import AzureOpenAI

client = AzureOpenAI(
    api_key="649c23fb56f142e99b225b0d27a489ab",
    api_version="2025-02-01-preview",
    azure_endpoint="https://hkust.azure-api.net"
)

response = client.chat.completions.create(
    model="gpt-5-mini",
    messages=[
        {"role": "system", "content": "You are a helpful assistant."},
        {"role": "user", "content": "What is Fibonacci sequence?"}
    ]
)
print(response.choices[0].message.content)
```

# INDEX

- Existing AI services
- Assignments

# IMAGE GENERATION SERVICES

Image generation models:

😄 High-quality, production-ready results, 😬 inflexibility - community

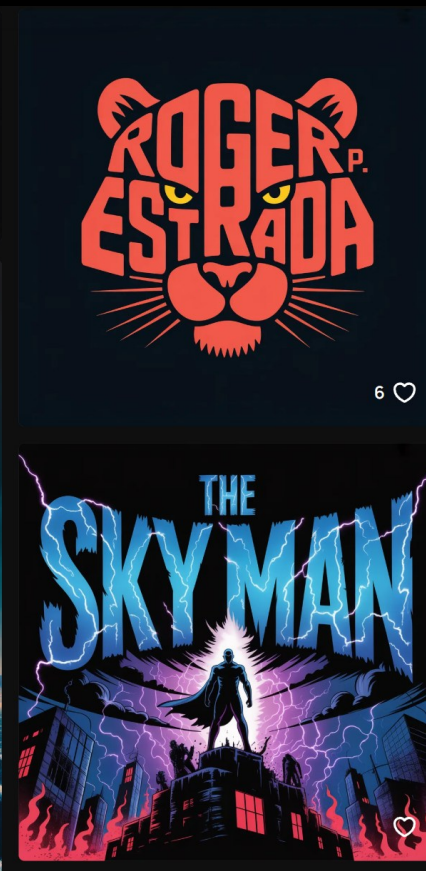
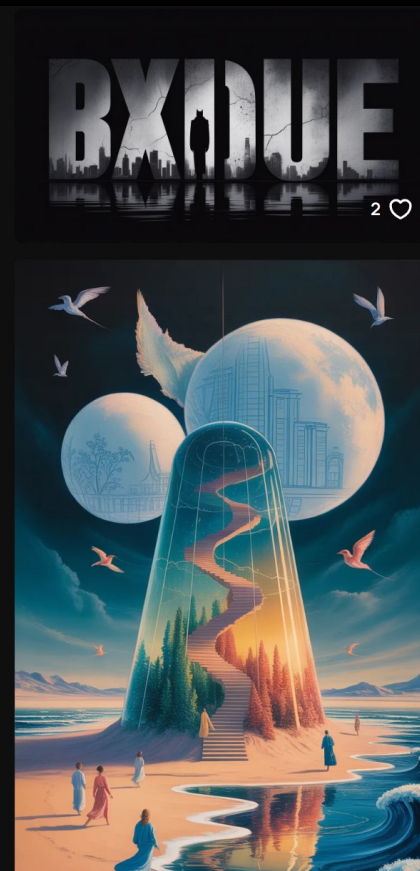
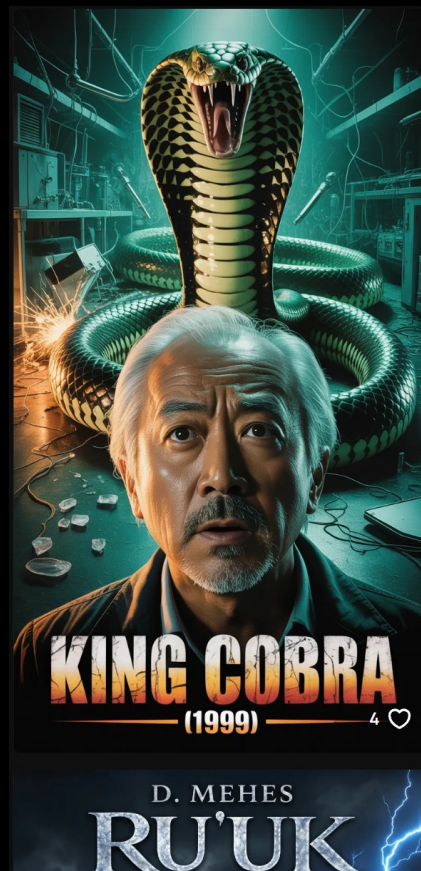
- Midjourney: <https://www.midjourney.com/explore?tab=hot> - The **best-looking** generation
- Ideogram : <https://ideogram.ai/login> -

Open-source image generation models:

😄 Flexible, large community, 😬 Require some efforts to make it work

- StableDiffusion: <https://stability.ai/stable-image>
- FLUX : <https://www.flux-ai.net/>
- Z-Image (from Alibaba Tongyi Lab): <https://github.com/Tongyi-MAI/Z-Image>

# MIDJOURNEY V.S. IDEOGRAM



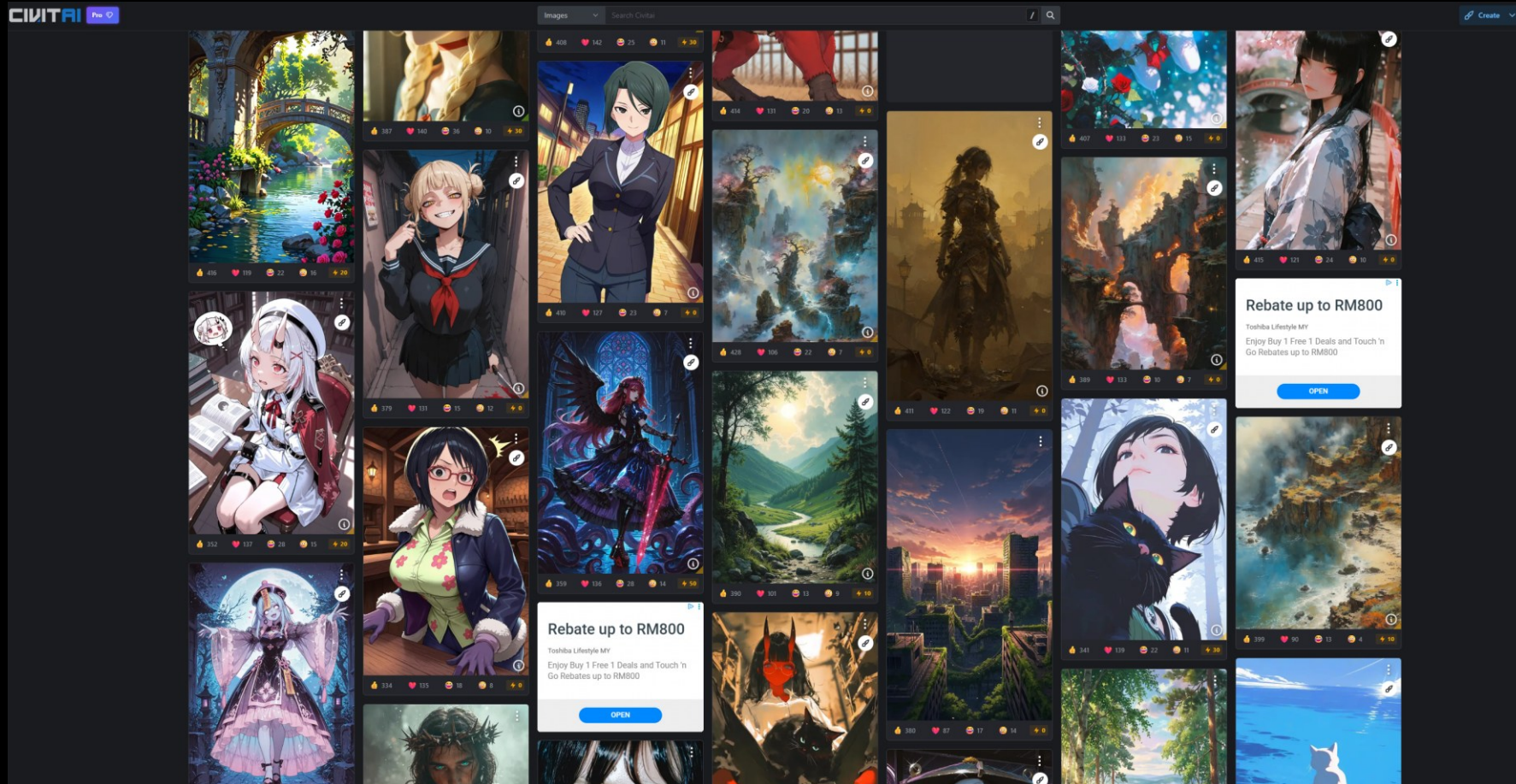
# MIDJOURNEY V.S. IDEOGRAM

| Feature             | Midjourney (v6 / v7 Alpha)                                        | Ideogram (v3.0)                                  |
|---------------------|-------------------------------------------------------------------|--------------------------------------------------|
| Primary Strength    | Cinematic photorealism and artistic interpretation                | Flawless text rendering and layout adherence     |
| Interface           | Discord-based (Web UI rolling out to more users)                  | Intuitive Web UI with a built-in "Canvas" editor |
| Typography Accuracy | Improved, but still hit-or-miss                                   | Near perfect                                     |
| Prompt Adherence    | Interprets prompts artistically (may ignore strict spatial logic) | Executes prompts literally and structurally      |



# OPEN SOURCE MODEL

- Strength lies in the community!



<https://civitai.com>



# LANGUAGE MODEL WITH IMAGE GEN

LLM + Image Generation – better instruction understanding

- DALL-E 3: <https://openai.com/index/dall-e-3/> - via HKUST Azure API
- Nano Banana: <https://gemini.google/overview/image-generation/>
- Qwen-Image: <https://github.com/QwenLM/Qwen-Image>
- Bit-Dance: <https://github.com/shallowdream204/BitDance>

# ASSIGNMENT 2 - CARD DESIGN

- Task 1: The Futuristic IP Character
  - Use Midjourney tools to design a collectible card featuring an original character/mascot.
  - **Theme:** Futuristic / Sci-Fi / Cyberpunk.
  - Introduce the concept of futuristic.
- Task 2: The Chinese New Year Redesign
  - Use Midjourney tools (e.g., Remix, Image Prompt) to transform your Futuristic IP into a **Chinese New Year** style
  - The core character is supposed to be recognizable.
- Submit:
  - Worksheet in PDF format
  - Slides to introduce your works (IP and Chinese New Year style).
  - Due on 5<sup>th</sup> March.
- Task 3: Present your work on the lecture session on 9<sup>th</sup> March.
  - You are only allowed to use the submitted slides to present your work (2-3 mins each)
  - Vote for your favourite works – due on 9<sup>th</sup> March.

# A SIMPLE & NAIVE EXAMPLE



For the IP character,  
What kind of characteristic?  
What kind of futuristic element?

...

For the CNY style,  
What does the character is doing?  
Why is this connected to CNY?

...

Any designs you think that are attractive!

# GRADING

- Grading Criteria (10 Points)
  - Finish all the tasks (1 pt)
  - Technical Execution (3 pts)  
Effective use of parameters/tools; consistent character identity; minimal AI artifacts ...
  - Aesthetics & Theme (3 pts)  
Visual appeal, composition, and accurate representation of both themes ...
- Peer Voting (3 pts)
  - Voting : 1 pt — If you submit your votes on time.
  - Rules: You have exactly 3 votes. You **cannot** vote for your own work.
  - Scoring: \* Rank 1-4: 2 pts  
Rank 5-8: 1 pt  
Rank 9-11: 0 pts